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5. Density

The density of liquid alkyl halides are higher than these of hydrocarbons of comparable molecular weight.

Evaluate Yourself

3) Write all possible chain isomers with molecular formula $C_5H_{11}Cl$

14.3.5 Chemical properties

Haloalkanes are one of the most reactive classes of organic compounds. Their reactivity is due to the presence of polar carbon – halogen bond in their molecules. The reactions of haloalkane may be divided into the following types

i) Nucleophilic substitution reactions

ii) Elimination reactions

iii) Reaction with metals

iv) Reduction

1) Nucleophilic substitution reactions

We know that the $C^{\delta+}$ - $X^{\delta-}$ present in halo alkane is polar and hence the nucleophilic reagents are attracted by partially positively charged carbon atoms resulting in substitution reactions.

Reaction with aqueous alkali or moist silver oxide.(Hydrolysis)

Haloalkane reacts with aqueous solution of KOH or moist silver oxide (Ag_2O/H_2O) to form alcohols.

Example

$CH_3 - CH_2 - Br + KOH\ (aq)$
Bromoethane
↓ boil
$CH_3 - CH_2 - OH + KBr$
Ethyl alcohol

$CH_3 - CH_2 - Br + AgOH$
Bromoethane
Moist silver oxide
↓ boil
$CH_3 - CH_2 - OH + AgBr$
Ethanol

i) Reaction with alcoholic ammonia (Ammonolysis)

Haloalkanes react with alcoholic ammonia solution to form alkyl amines.

Example

$CH_3 - CH_2 - Br + H - NH_2$
Bromo ethane ↓ ammonia

$CH_3 - CH_2 - NH_2 + HBr$
Ethyl amine

However, with excess of halo alkane, secondary and tertiary amines along with quartenary ammonium salts are obtained

$CH_3CH_2NH_2 \xrightarrow[-HBr]{CH_3CH_2Br} (CH_3CH_2)_2NH$
Ethyl amine (1)
Diethyl amine (2)
↓ CH_3CH_2Br -HBr
$(CH_3CH_2)_3N$
Triethyl amine (3)
↓ CH_3CH_2Br -HBr
$(CH_3CH_2)_4N^+Br^-$
Tetra ethyl ammonium bromide